Natural Language Processing

Class 8: Sequence Labeling Tasks

Adam Faulkner

March 17, 2026

1. Part-of-Speech Tagging
2. Syntactic Parsing
3. Named Entity Recognition
4. Discourse Coherence

1 Part-of-Speech Tagging

2 Syntactic Parsing

3 Named Entity Recognition

4 Discourse Coherence

Parts of Speech

From the earliest linguistic traditions (Yaska and Panini 5th C. BCE, Aristotle 4th C. BCE), the idea that words can be classified into grammatical categories

- part of speech, word classes, POS, POS tags 8 parts of speech attributed to Dionysius Thrax of Alexandria (c. 1st C. BCE):
- noun, verb, pronoun, preposition, adverb, conjunction, participle, article
- These categories are relevant for NLP today

Part-of-Speech Tagging

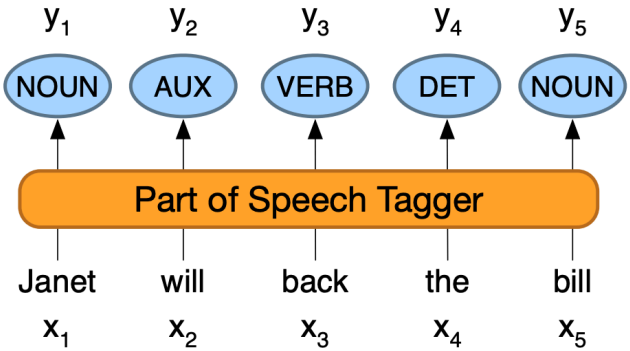
- Assign a part-of-speech to each word in a text.
- Words often have more than one POS.

book:

- VERB: (**Book** that flight)
- NOUN: (Hand me that **book**).

Part-of-Speech Tagging

Map from sequence $x_1,\ldots,x_n$ of words to $y_1,\ldots,y_n$ of POS tags



Part-of-Speech Tagging

There/PRO were/VERB 70/NUM children/NOUN there/ADV ./PUNC

Preliminary/ADJ findings/NOUN were/AUX reported/VERB in/ADP today/NOUN 's/PART New/PROPN England/PROPN Journal/PROPN of/ADP Medicine/PROPN

How difficult is POS tagging in English?

- Roughly 15% of word types are ambiguous
- Hence 85% of word types are unambiguous
  - *Janet* is always **PROPN**, *hesitantly* is always **ADV**
- But those 15% tend to be very common.
- So 60% of word tokens are ambiguous
- E.g., **back**
  - *earnings growth took a **back/ADJ** seat*
  - *a small building in the **back/NOUN***
  - *a clear majority of senators **back/VERB** the bill*
  - *enable the country to buy **back/PART** debt*
  - *I was twenty-one **back/ADV** then*

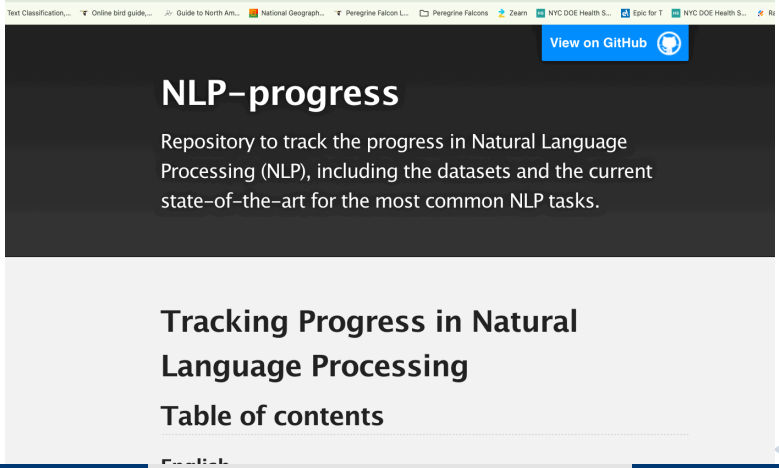
Standard algorithms for POS tagging

Supervised Machine Learning Algorithms:

- Hidden Markov Models
- Conditional Random Fields (CRF) / Maximum Entropy Markov Models (MEMM)
- Neural sequence models (RNNs or Transformers)
- Large Language Models (like BERT), finetuned
- All required a hand-labeled training set, all about equal performance (97% on English)
  - Via human created features: HMMs and CRFs
  - Via representation learning: Neural LMs

State-of-the-Art Approaches to Part-of-Speech Tagging

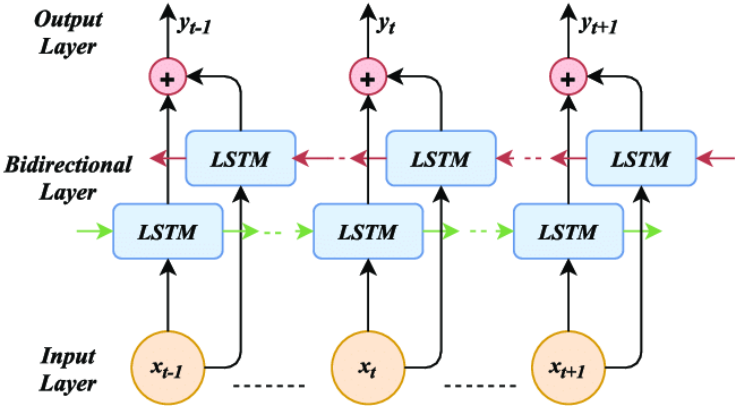
A handy resource for quickly checkinge the SoTA for POS-tagging and many other NLP tasks is Sebastian Ruder's `https://nlpprogress.com/` site



Part-of-Speech Tagging using Bidirectional LSTMs

- Bidirectional LSTMs (BiLSTM) and finetuned BERT-based models dominate SoTA approaches to POS-tagging (and many other sequence modeling tasks such as NER)
- BiLSTMs process sequential data in both forward and reverse temporal directions
- At each position in the input sequence, the outputs from the forward and backward LSTM traversal are concatenated, providing a rich context that incorporates both past and future information
- Effective for tasks where context on both sides of a token or time step is critical

Part-of-Speech Tagging using Bidirectional LSTMs



1 Part-of-Speech Tagging

2 Syntactic Parsing

3 Named Entity Recognition

4 Discourse Coherence

Chomsky's argument against probabilistic models of language: A key chapter in the history of AI

Noam Chomsky (1957)

"It must be recognized that the notion 'probability of a sentence' is an entirely useless one, under any known interpretation of this term."

Chomsky's famous example of a grammatically correct but 0 probability sentence

Noam Chomsky (1957)

"Colorless green ideas sleep furiously."

- **Observation 1:** The sentence is nonsensical (semantically null).
- **Observation 2:** It is intuitively recognized by English speakers as **grammatically correct**.
- **Contrast:** Compare with *"Furiously sleep ideas green colorless."* which is both nonsensical and ungrammatical.

Chomsky's famous example of a grammatically correct but 0 probability sentence

Noam Chomsky (1957)

"Colorless green ideas sleep furiously."

- **Observation 1:** The sentence is nonsensical (semantically null).
- **Observation 2:** It is intuitively recognized by English speakers as **grammatically correct**.
- **Contrast:** Compare with *"Furiously sleep ideas green colorless."* which is both nonsensical and ungrammatical.

Chomsky's Argument Against Probability

Chomsky argued that Markov chains or N-gram models fail because:

1. **Statistical Rarity:** In 1957, both "Colorless green..." and its reversal had a probability of essentially 0 in any corpus.
2. **Independence of Syntax:** If both sentences have $P = 0$, a probabilistic model cannot distinguish between the "grammatical" one and the "ungrammatical" one.
3. **Conclusion:** Knowledge of language is based on abstract rules (syntax), not on the frequency of word sequences.

The Counter-Argument: Fernando Pereira (2000)

Pereira showed that statistical models **can** distinguish these sentences if given enough data and better smoothing.

- Ignores smoothing methods (which were around even when Chomsky provided his example). Pereira: "To avoid this, one usually smoothes the data, using a lower-degree polynomial that may not fit the training data exactly but that will be less dependent on the vagaries of the sample. Similarly, smoothing methods can be used in probability models to assign some probability mass to unseen events"
- Using a hidden Markov model trained on newspaper text, Pereira showed that "Colorless green ideas sleep furiously" was **200,000 times more likely** than its reverse.
- **Key Insight:** Probability doesn't just track exact strings; it tracks the underlying patterns of the language.

Why do we need sentence structure?

- Humans communicate complex ideas by composing words together into bigger units to convey complex meanings
- Listeners need to work out what modifies [attaches to] what
- A model needs to understand sentence structure in order to be able to interpret language correctly

Constituency Grammar (CG) Review

- The traditional approach to syntax (Phrase Structure Grammar).
- Focuses on constituents (phrases) that act as single units.
- Structure is represented by a tree with non-terminal (phrase) nodes (NP, VP, S).

Constituency Grammar (CG) Review

- The traditional approach to syntax (Phrase Structure Grammar).
- Focuses on constituents (phrases) that act as single units.
- Structure is represented by a tree with non-terminal (phrase) nodes (NP, VP, S).

The Core Limitation

CGs focus on grouping words, often obscuring the direct relationships (who does what to whom).They don't explicitly model the head-dependent relationship.

Constituency Structure (phrase structure)

- Syntactic constituency is the idea that groups of words can behave as single units, or constituents. Part of developing a grammar involves building an inventory of the constituents in the language.
- Phrase structure organizes words into nested constituents.
- How do we know what a constituent is?
- Distribution: A constituent behaves as a unit that can appear in different places:
  - John talked [to the children] [about drugs].
  - John talked [about drugs] [to the children].
  - *John talked drugs to the children about

  Substitution/expansion/pronoun replacement
  - I sat [on the box/right on top of the box/there].

Chomsky, 1956: Context-Free Grammars

- A context-free grammar consists of a set of rules or productions, each of which expresses the ways that symbols of the language can be grouped and ordered together, and a lexicon of words and symbols.
- For example, the following productions NP express that an NP (or noun phrase) can be composed of either a ProperNoun or a determiner (Det) followed by a Nominal; a Nominal in turn can consist of one or more Nouns.

- NP → Det Nominal
- NP → ProperNoun
- Nominal → Noun | Nominal Noun

Chomsky, 1956: Context-Free Grammars

- Context-free rules can be hierarchically embedded, so we can combine the previous rules with others, like the following, that express facts about the lexicon:
  - Det → a
  - Det → the
  - Noun → flight
- Terminals: The symbols that correspond to words in the language (the, nightclub); the lexicon is the set of rules that introduce these terminal symbols.
- Non-terminals: In each context-free rule, the item to the right of the arrow (→) is an ordered list of one or more terminals and non-terminals; to the left of the arrow is a single non-terminal

Chomsky, 1956: Context-Free Grammars

$G = (T, N, S, R)$

- T is a set of terminal symbols
- N is a set of nonterminal symbols
- S is the start symbol ($S \in N$)
- R is a set of rules/productions of the form $X \rightarrow \gamma$
  - $X \in N$ and $\gamma \in (N \cup T)^*$

A grammar G generates a language L.

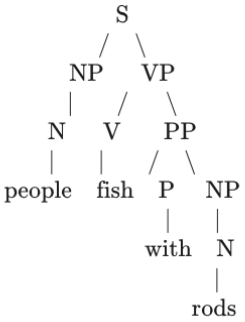
Given the following CFG...

S → NP VP	N → people
VP → V NP	N → fish
VP → V NP PP	N → tanks
NP → NP NP	N → rods
NP → NP PP	V → people
NP → N	V → fish
NP → *e*	V → tanks
PP → P NP	P → with

people fish tanks

people fish with rods

...we can generate a parse for "people fish with rods"



We generated this parse using

- S → NP VP
- VP → V PP
- PP → P NP
- NP → N
- N → people, fish, rods
- P → with

Probabilistic context-free grammars (PCFGs)

G = (T, N, S, R, P)

- T is a set of terminal symbols
- N is a set of nonterminal symbols
- S is the start symbol (S $\in$ N)
- R is a set of rules/productions of the form X $\rightarrow \gamma$
- P is a probability function
  - P: R $\rightarrow$ [0,1]
  - $\forall X \in N, \sum_{X \rightarrow \gamma \in R} P(X \rightarrow \gamma) = 1$

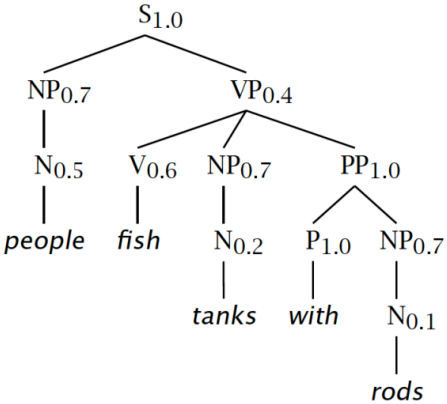
A grammar G generates a language model L.

$$\sum_{\gamma \in T*} P(\gamma) = 1$$

The probability of trees and strings

- $P(t)$ The probability of a tree t is the product of the probabilities of the rules used to generate it.
- $P(s)$ The probability of the string s is the sum of the probabilities of the trees that have that string as their yield

$$P(s) = \sum_j P(s,t), \text{ where t is a parse of s} = \sum_j P(t)$$


S1.0
NP0.7
VP0.4
N0.5
V0.6
NP0.7
PP1.0
people
fish
N0.2
P1.0
NP0.7
tanks
with
N0.1
rods

Evaluating Parsers

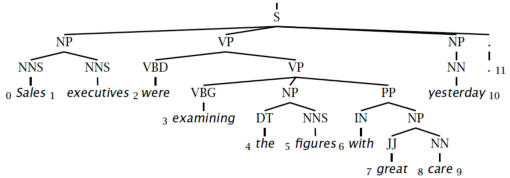
- How much do the constituents in the hypothesis parse tree look like the constituents in a hand-labeled, reference parse?
- Requires a human-labeled reference (or "gold standard") parse tree for each sentence in the test set
- A constituent in a hypothesis parse of a sentence is labeled correct if there is a constituent in the reference parse with the same starting point, ending point, and non-terminal symbol.

$$\textbf{labeled recall:} = \frac{\text{\# of correct constituents in hypothesis parse of } s}{\text{\# of total constituents in reference parse of } s}$$

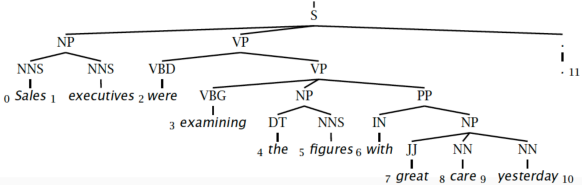
$$\textbf{labeled precision:} = \frac{\text{\# of correct constituents in hypothesis parse of } s}{\text{\# of total constituents in hypothesis parse of } s}$$

Evaluating Parsers)

Gold standard brackets: **S-(0:11)**, **NP-(0:2)**, VP-(2:9), VP-(3:9), **NP-(4:6)**, PP-(6-9), NP-(7,9), NP-(9:10)



Candidate brackets: **S-(0:11)**, **NP-(0:2)**, VP-(2:10), VP-(3:10), **NP-(4:6)**, PP-(6-10), NP-(7,10)



Evaluating Parsers)

Gold standard brackets:

S-(0:11), NP-(0:2), VP-(2:9), VP-(3:9), **NP-(4:6)**, PP-(6-9), NP-(7,9), NP-(9:10)

Candidate brackets:

S-(0:11), **NP-(0:2)**, VP-(2:10), VP-(3:10), **NP-(4:6)**, PP-(6-10), NP-(7,10)

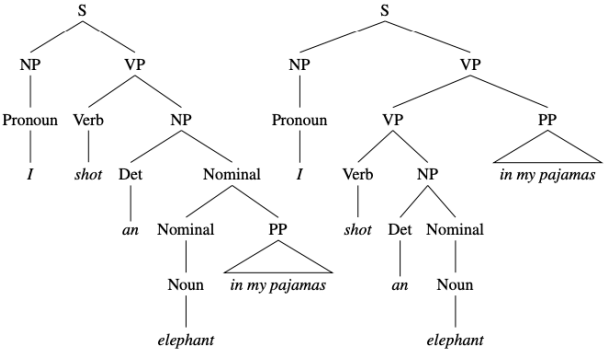
Labeled Precision	3/7 = 42.9%
Labeled Recall	3/8 = 37.5%
LP/LR F1	40.0%
Tagging Accuracy	11/11 = 100.0%

How parse trees resolve different kinds of ambiguity

- Ambiguity is the most serious problem faced by syntactic parsers.
- **Structural ambiguity** occurs when the grammar can assign more than one parse to a sentence.
- Groucho Marx: *One morning* ***I shot an elephant in my pajamas****. How he got into my pajamas I'll never know*

How parse trees resolve different kinds of ambiguity

Ambiguous because the phrase *in my pajamas* can be part of the NP headed by *elephant* or a part of the verb phrase headed by *shot*



Treebanks

- A corpus in which every sentence is annotated with a parse tree is called a treebank
- Treebanks are generally made by running a parser over each sentence and then having the resulting parse hand-corrected by human linguists.
- The most well-known of these is the **Penn Treebank**

```
((S
   (NP-SBJ (DT That)
     (JJ cold) (, ,)
     (JJ empty) (NN sky) )
   (VP (VBD was)
     (ADJP-PRD (JJ full)
       (PP (IN of)
         (NP (NN fire)
           (CC and)
           (NN light) ))))
   (. .) ))
```

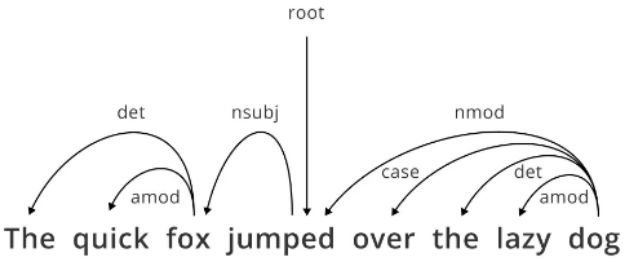
(a)

```
((S
   (NP-SBJ The/DT flight/NN )
   (VP should/MD
     (VP arrive/VB
       (PP-TMP at/IN
         (NP eleven/CD a.m/RB ))
       (NP-TMP tomorrow/NN )))))
```

(b)

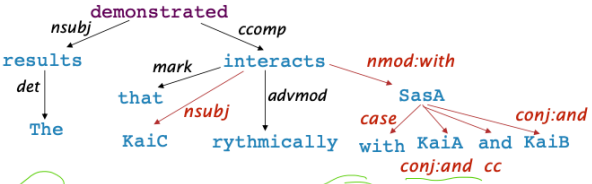
Alternative to constituency structure: dependency structure

- Dependency syntax postulates that syntactic structure consists of relations between lexical items, normally binary asymmetric relations (arrows) called dependencies
- Dependency structure shows which words depend on (modify, attach to, or are arguments of) which other words.



Alternative to constituency structure: dependency structure

- Dependency paths help extract semantic interpretation simple practical example: extracting protein-protein interaction



KaiC ←nsubj interacts nmod:with → SasA
KaiC ←nsubj interacts nmod:with → SasA conj:and→ KaiA
KaiC ←nsubj interacts nmod:with → SasA conj:and→ KaiB

1 Part-of-Speech Tagging

2 Syntactic Parsing

3 Named Entity Recognition

4 Discourse Coherence

Named Entities

- Named entity, in its core usage, means anything that can be referred to with a proper name. Most commonly used 4 tags:
  - PER (Person): “Marie Curie”
  - LOC (Location): “New York City”
  - ORG (Organization): “Stanford University”
  - GPE (Geo-Political Entity): “Boulder, Colorado”
- Often multi-word phrases
- But the term is also extended to things that aren’t entities: dates, times, prices

Named Entity tagging

The task of named entity recognition (NER):

- Find spans of text that constitute proper names
- Tag the type of entity

NER output

Citing high fuel prices, [$_{ORG}$ **United Airlines**] said [$_{TIME}$ **Friday**] it has increased fares by [$_{MONEY}$ **$6**] per round trip on flights to some cities also served by lower-cost carriers. [$_{ORG}$ **American Airlines**], a unit of [$_{ORG}$ **AMR Corp.**], immediately matched the move, spokesman [$_{PER}$ **Tim Wagner**] said. [$_{ORG}$ **United**], a unit of [$_{ORG}$ **UAL Corp.**], said the increase took effect [$_{TIME}$ **Thursday**] and applies to most routes where it competes against discount carriers, such as [$_{LOC}$ **Chicago**] to [$_{LOC}$ **Dallas**] and [$_{LOC}$ **Denver**] to [$_{LOC}$ **San Francisco**].

Why NER?

- Sentiment analysis: Consumers sentiment toward a particular company or person
- Question Answering: Answer questions about which entity in a given sentence?
- Information Extraction: Extracting facts about entities from text.

Why NER is hard

- Segmentation: In POS tagging, no segmentation problem since each word gets one tag.
- In NER we have to find and segment the entities!
- Type ambiguity:

[$_{\text{PER}}$ Washington] was born into slavery on the farm of James Burroughs.
[$_{\text{ORG}}$ Washington] went up 2 games to 1 in the four-game series.
Blair arrived in [$_{\text{LOC}}$ Washington] for what may well be his last state visit.
In June, [$_{\text{GPE}}$ Washington] passed a primary seatbelt law.

BIO tagging scheme

- How can we turn this structured problem into a sequence problem like POS tagging, with one label per word?

[PER Jane Villanueva] of [ORG United] , a unit of [ORG United Airlines Holding] , said the fare applies to the [LOC Chicago] route.

BIO tagging scheme

[PER Jane Villanueva] of [ORG United] , a unit of [ORG United Airlines Holding] , said the fare applies to the [LOC Chicago] route.

Words	BIO Label
Jane	B-PER
Villanueva	I-PER
of	O
United	B-ORG
Airlines	I-ORG
Holding	I-ORG
discussed	O
the	O
Chicago	B-LOC
route	O
.	O

BIO tagging scheme

Words	BIO Label
Jane	B-PER
Villanueva	I-PER
of	O
United	B-ORG
Airlines	I-ORG
Holding	I-ORG
discussed	O
the	O
Chicago	B-LOC
route	O
.	O

- **B**: token that begins a span
- **I**: tokens inside a span
- **O**: tokens outside of any span
- # of tags (where n is entity types):
  - 1 O tag,
  - n B tags,
  - n I tags
  - total of 2n+1

BIO Tagging variants: IO and BIOES

[PER Jane Villanueva] of [ORG United] , a unit of [ORG United Airlines Holding] , said the fare applies to the [LOC Chicago] route.

Words	IO Label	BIO Label	BIOES Label
Jane	I-PER	B-PER	B-PER
Villanueva	I-PER	I-PER	E-PER
of	O	O	O
United	I-ORG	B-ORG	B-ORG
Airlines	I-ORG	I-ORG	I-ORG
Holding	I-ORG	I-ORG	E-ORG
discussed	O	O	O
the	O	O	O
Chicago	I-LOC	B-LOC	S-LOC
route	O	O	O
.	O	O	O

Standard algorithms for NER

Supervised Machine Learning:Needs a human-labeled training set of text annotated with tags

- Hidden Markov Models
- Conditional Random Fields (CRF)/ Maximum Entropy Markov Models (MEMM)
- Neural sequence models (RNNs or Transformers) Large Language Models (like BERT), finetuning (the accompanying Jupyter notebook provides an example)

1 Part-of-Speech Tagging

2 Syntactic Parsing

3 Named Entity Recognition

4 Discourse Coherence

What is Discourse Coherence?

- **Discourse Coherence** refers to the quality of a text or spoken conversation that makes it **semantically unitary** and **meaningful**.
- It is the way in which the sentences and ideas in a text relate to each other to form a whole.
- A coherent text is not just a collection of random sentences; it has a logical flow and structure.
- Coherence is essential for comprehension and interpretation.
- John took a train from Paris to Istanbul. He likes spinach. vs *Jane took a train from Paris to Istanbul. She had to attend a conference*

Types of Coherence

- **Local Coherence**
  - Deals with the relationship between **adjacent sentences** or clauses.
  - Focuses on immediate links and transitions.
  - Often achieved through **cohesion** (lexical and grammatical ties).
- **Global Coherence**
  - Deals with the relationship of all sentences to the **overall topic** or goal of the discourse.
  - Ensures the entire text makes sense as a whole and follows a central theme (e.g., narrative plot, argument structure).

Lexical Cohesion

- Refers to the coherence created by the **choice of words** and the semantic relationships between them.
- Achieved through:
  - **Repetition**: Repeating the same word or phrase.
  - **Synonymy/Antonymy**: Using words with similar or opposite meanings.
  - **Hyponymy/Hypernymy**: Using words that are sub-classes or super-classes (e.g., *dog* is a hyponym of *animal*).
  - **Collocation**: Words that frequently occur together (e.g., *heavy rain*).

Entity-Based Coherence

- Focuses on the **tracking of entities** (people, objects, concepts) throughout the text.
- Achieved primarily through **reference resolution**:
  - **Anaphora**: Using a pronoun or shorter phrase to refer back to an already introduced entity (e.g., *John* arrived. *He* sat down).
  - Coherence is maintained if the reader can easily identify the entities being discussed and how they are referred to in subsequent sentences.

Coherence Relations (Discourse Relations)

- These are the **explicit or implicit links** that hold two segments of text (propositions, clauses, sentences) together.
- They describe **how the content of one segment relates to the content of another**.
- Examples of common relations:
  - **Causality**: (Because, Since)
  - **Contrast**: (However, But)
  - **Elaboration**: (Specifically, For example)
  - **Temporal**: (Then, After)
  - **Condition**: (If, Provided that)

Rhetorical Structure Theory (RST)

- A widely used theory for modeling discourse structure, developed by Mann and Thompson (1988).
- Models the text as a **tree structure** where non-overlapping spans of text are linked by **coherence relations**.
- The basic structure involves:
  - **Nucleus**: The central, essential part of the relation.
  - **Satellite**: The subsidiary, less central part (often provides background, evidence, or context).
- **Goal**: Explain the coherence of a text by identifying its components and the rhetorical relations that hold between them.

Key Concepts in RST

- **Relation:** The link that connects two spans of text.
- **Nucleus (NUC):** The essential, central part of the relation. The speaker's primary intent is often conveyed here.
- **Satellite (SAT):** The subsidiary, less essential part. It supports, elaborates, or provides context for the Nucleus.

1. Reason

- **Definition:** The nucleus is an action carried out by an animate agent and the satellite is the **reason** for the nucleus.
- **Goal:** To explain the motivation for the action in the Nucleus.

`[NUC Jane took a train from Paris to Istanbul.] [SAT She had to attend a conference.]`

- The action (taking the train) is the NUC.
- The motivation (attending a conference) is the SAT.

2. Elaboration

- **Definition:** The satellite gives **additional information or detail** about the situation presented in the nucleus.
- **Goal:** To provide richness, specificity, or context to the Nucleus.

`[NUC Dorothy was from Kansas.] [SAT She lived in the midst of the great Kansas prairies.]`

- The core fact (being from Kansas) is the NUC.
- The detailed description (where she lived) is the SAT.

3. Evidence

- **Definition:** The satellite gives information with the goal of **convincing the reader** to accept the information presented in the nucleus.
- **Note:** Unlike `Reason`, which explains an agent's motivation, `Evidence` supports the **belief** in a proposition.

`[NUC Kevin must be here.] [SAT His car is parked outside.]`

- The claim (Kevin is here) is the NUC.
- The supporting observation (car parked outside) is the SAT.

4. Attribution

- **Definition:** The satellite gives the **source of attribution** for an instance of reported speech or thought in the nucleus.
- **Goal:** To credit the source of the reported information.

`[SAT Analysts estimated] [NUC that sales at U.S. stores declined in the quarter, too]`

- The reported content (sales declined) is the NUC.
- The source of the report (Analysts estimated) is the SAT.
- **Note:** The SAT precedes the NUC in this example.

5. List

- **Definition:** A **multinuclear** relation where a series of nuclei is given, without contrast or explicit comparison.
- **Note:** In multinuclear relations, **all spans are Nuclei** because they are all central to the writer's purpose.

```
[NUC Billy Bones was the mate; ] [NUC Long John, he was quartermaster]
```

- Both items (Billy Bones' role and Long John's role) are equally important and are merely listed as related facts.

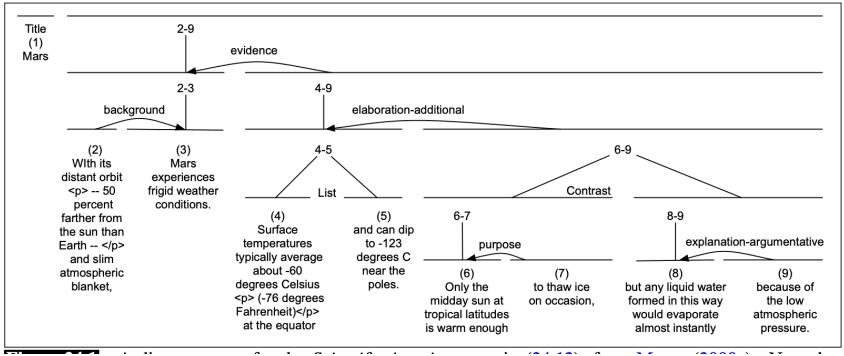
Summary of Relations

Relation	Function of Satellite
Reason	Explains the **motive** for an action
Elaboration	Provides **detail** about the NUC
Evidence	Supports the **belief** in the NUC
Attribution	Gives the **source** of the NUC's content
List	Presents **equally weighted items** (Multinuclear)

Example RST parse: Scientific American Article

(24.12) With its distant orbit–50 percent farther from the sun than Earth–and slim atmospheric blanket, Mars experiences frigid weather conditions. Surface temperatures typically average about -60 degrees Celsius (-76 degrees Fahrenheit) at the equator and can dip to -123 degrees C near the poles. Only the midday sun at tropical latitudes is warm enough to thaw ice on occasion, but any liquid water formed in this way would evaporate almost instantly because of the low atmospheric pressure.

Example RST parse: Scientific American Article



The Penn Discourse Treebank (PDTB)

- A large-scale corpus annotated with **discourse relations**.
- Distinctive features:
  - Annotation is **local**: Relations are marked only between adjacent arguments.
  - Focuses on **connectives** (e.g., *although, because, and*) and implicit relations.
  - **Arguments** of a relation are entire spans of text (e.g., sentence or clause).
  - **Lexicalized approach**: Relations are anchored to a discourse connective (explicit or implicit).
- PDTB is crucial for training and evaluating automated discourse parsers.

Discourse Parsing

- The computational task of automatically **determining the structure and coherence relations** in a text.
- It is a form of higher-level linguistic analysis, going beyond syntactic parsing.
- **Goal**: To represent the discourse structure explicitly for natural language processing (NLP) applications like:
  - Text Summarization (identifying the central 'Nucleus' parts).
  - Machine Translation (handling cross-sentential dependencies).
  - Question Answering (understanding causal links).
- Parsers often rely on machine learning models trained on corpora like RST-DT (for RST) or the PDTB.

Next class: March 24

Assignment 3: Due March 24

Text generation tasks

Reading

- Jurafsky & Martin Chapter 12: Machine Translation
- A Systematic Survey of Text Summarization: From Statistical Methods to Large Language Models
- Paraphrase Generation: A Survey of the State of the Art
- Commonsense Knowledge Reasoning and Generation with Pre-trained Language Models: A Survey